EXPERIMENT 9:

AIM:

To create a Pivot Table using Pandas and calculate the total sale amount region-wise, manager-wise, and salesman-wise.

INPUT:

Region,Manager,Salesman,Item,Sale_Amount

North,John,Ravi,Laptop,50000

North,John,Amit,Printer,15000

South,Raj,Kumar,Laptop,45000

South,Raj,Anil,Mouse,5000

East,Peter,Suresh,Monitor,18000

East,Peter,Ramesh,Keyboard,7000

West,David,Arun,Laptop,55000

West,David,Vijay,Printer,16000

North,John,Ravi,Keyboard,6000

South,Raj,Kumar,Monitor,20000

PROGRAM:

exp9.py > ...

```python
import pandas as pd
df = pd.read_csv(r"C:\Users\dhars\Documents\Query processing\lab_questions\exp9.csv")
print("Original Data")
print(df)
pivot_table = pd.pivot_table(
    df,
    values='Sale_Amount',
    index=['Region', 'Manager', 'Salesman'],
    aggfunc='sum'
)
print("\nPivot Table")
print(pivot_table)
```

OUTPUT:

```
\Python313\python.exe c:/Users/dhars/Documents/Qu
Original Data
  Region Manager Salesman      Item  Sale_Amount
0  North    John     Ravi    Laptop        50000
1  North    John     Amit   Printer        15000
2  South     Raj    Kumar    Laptop        45000
3  South     Raj     Anil     Mouse         5000
4   East   Peter   Suresh   Monitor        18000
5   East   Peter   Ramesh  Keyboard         7000
6   West   David     Arun    Laptop        55000
7   West   David    Vijay   Printer        16000
8  North    John     Ravi  Keyboard         6000
9  South     Raj    Kumar   Monitor        20000

Pivot Table
                         Sale_Amount
Region Manager Salesman
East   Peter   Ramesh           7000
               Suresh          18000
North  John    Amit            15000
               Ravi            56000
South  Raj     Anil             5000
               Kumar           65000
West   David   Arun            55000
               Vijay           16000
```

RESULT:

Thus, the Pivot Table was successfully created and the total sale amount was calculated region-wise, manager-wise, and salesman-wise.

EXPERIMENT 10:

AIM:

To create a DataFrame with ten rows and four columns containing random values and highlight negative numbers in red and positive numbers in black using Pandas styling.

INPUT:

A,B,C,D

10,-20,35,-40

-15,25,-30,45

50,-60,70,-80

-90,100,-10,20

30,-40,50,-60

-70,80,-90,10

15,-25,35,-45

55,-65,75,-85

95,-5,15,-25

40,-50,60,-70

PROGRAM:

exp10.py > ...

```
import pandas as pd
df = pd.read_csv(r"C:\Users\dhars\Documents\Query processing\lab_questions\exp10.csv")
print("Original DataFrame")
print(df)
def colour(val):
    if val < 0:
        return "color: red"
    else:
        return "color: black"

styled_df = df.style.map(colour)
styled_df.to_html("output.html")
print("\nStyled DataFrame saved as output.html")
```

OUTPUT:

	A	B	C	D
0	10	-20	35	-40
1	-15	25	-30	45
2	50	-60	70	-80
3	-90	100	-10	20
4	30	-40	50	-60
5	-70	80	-90	10
6	15	-25	35	-45
7	55	-65	75	-85
8	95	-5	15	-25
9	40	-50	60	-70

RESULT:

Thus, a DataFrame with random values was successfully created, and the negative numbers were highlighted in red while the positive numbers were displayed in black using Pandas styling.